

Laboratory Manual

For

Basic Electrical Engineering Sessional
EEE-1122



Department of Computer Science and Engineering
IIUC

International Islamic University Chittagong (IIUC)
Department of Computer Science and Engineering

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EEE-1121: Basic Electrical Engineering Sessional

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International Islamic University Chittagong (IIUC)
Dept. of Computer Science and Engineering (CSE)
Course Code: EEE-1122 Course Title: Basic Electrical Engineering Sessional

Experiment No.: 01

Experiment Name: “Guidance and introduction of this course and the basic instruments, including remedial instruction of Basic Electronic Engineering Laboratory and Safety.”

Objectives:

This experiment will provide exposure to the various test equipment to be used in subsequent experiments. A primary purpose of this lab course is for you to teach the use of electrical and electronics test equipment. The devices we will be using include DC power supplies, breadboards, and digital multi-meters” DMM”, as well as some resistors, so we can have something to measure on. By the end of this experiment you should be able to tell how to use the equipment to measure voltage, current resistance, current and voltage.

Apparatus Required:

S no.	Particulars	Specification/Range	Quantity	Make/Model No.
1.				
2.				
3.				
4.				
5.				

Theory:

Digital multimeter

A multi-meter or a multi-tester, also known as a VOM (Volt-Ohm meter), is an electronic measuring instrument that combines several measurement functions in one unit. A typical multimeter may include features such as the ability to measure voltage, current and resistance. Multimeters may use analog or digital circuits— analog multimeters (AMM) and digital multimeters (often abbreviated DMM or DVOM.) Analog instruments are usually based on a microammeter whose pointer moves over a scale calibrated for all the different measurements that can be made; digital instruments usually display digits but may display a bar of a length proportional to the quantity being measured.

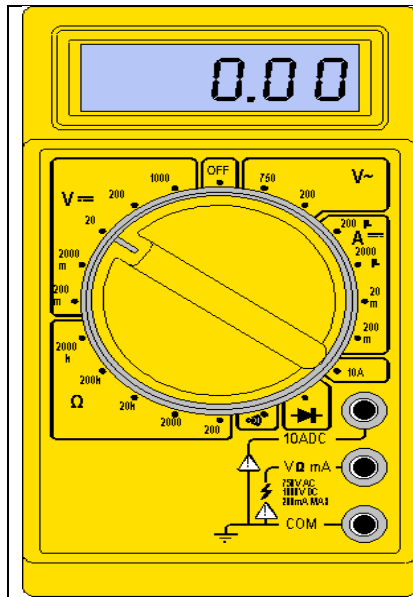


Fig: DMM

Quantities measured: Contemporary multimeters can measure many quantities. The common ones are:

- Voltage, alternating and direct, in volts.
- Current, alternating and direct, in amperes.
- The frequency range for which AC measurements are accurate must be specified.
- Resistance in ohms.
- Capacitance in farads.
- Frequency in hertz.

There are two types of multimeters

- 1) Analog multimeter
- 2) Digital multimeter

A meter is a measuring instrument. An ammeter measures current, a voltmeter measures the potential difference (voltage) between two points, and an ohmmeter measures resistance. A multimeter combines these functions and possibly some additional ones as well, into a single instrument.

Resistor and Resistance measurement using DMM:

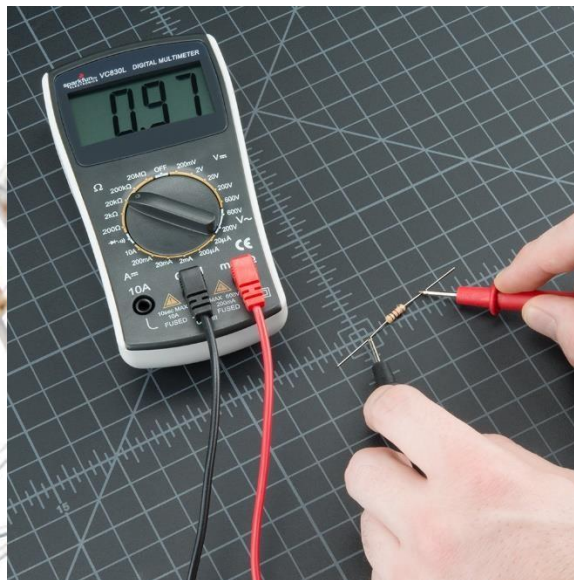
Resistors are electronic components which have a specific, never-changing electrical resistance. The resistor's resistance limits the flow of electrons through a circuit. They are passive components, meaning they only consume power (and can't generate it). Resistors are usually added to circuits where they complement active components like op-amps, microcontrollers, and other integrated circuits. Commonly resistors are used to limit current, divide voltages, and pull-up I/O lines.

Resistor units

The electrical resistance of a resistor is measured in ohms. The symbol for an ohm is the greek capital-omega: Ω . The (somewhat roundabout) definition of 1Ω is the resistance between two points where 1 volt (1V) of applied potential energy will push 1 ampere (1A) of current.



Resistor



resistance measurement using DMM

Ref: <https://learn.sparkfun.com/tutorials/resistors/all>

Breadboard:

A breadboard is used to make up temporary circuits for testing or to try out an idea. No soldering is required so it is easy to change connections and replace components. Parts are not damaged and can be re-used afterwards. The photograph shows a typical small breadboard which is suitable for beginners building simple circuits with one or two ICs (chips).

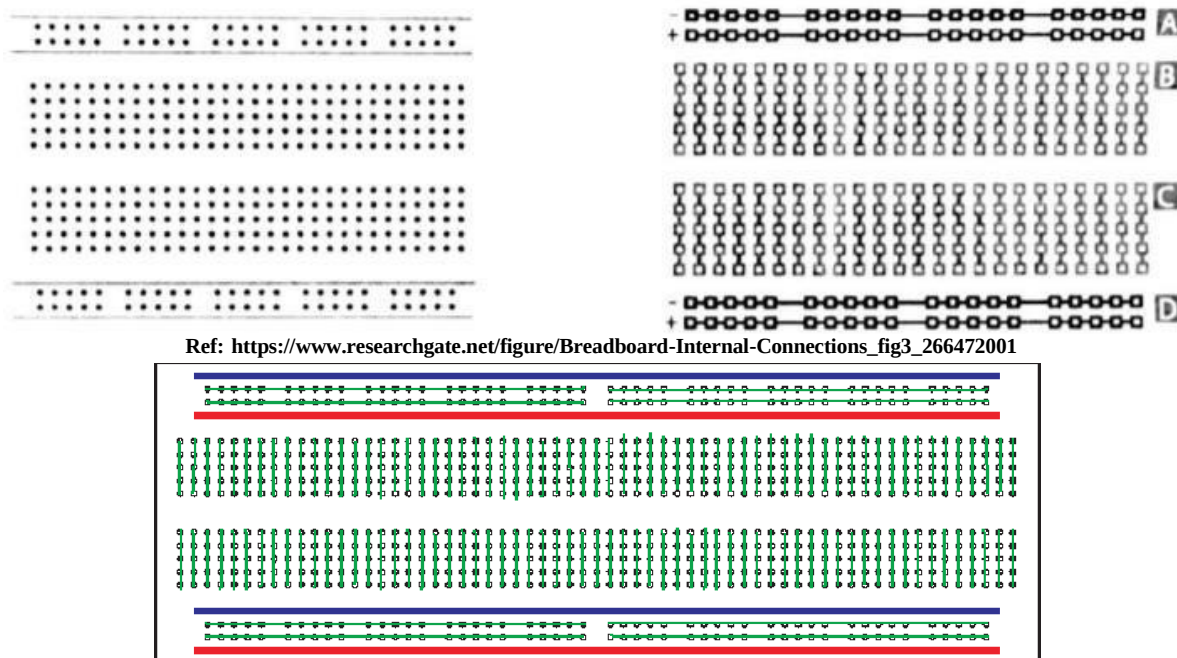
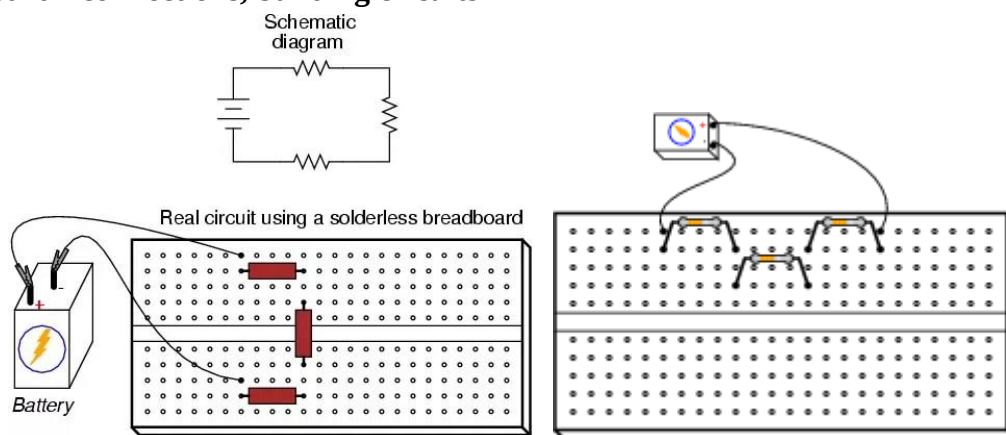


Fig: Breadboard

Breadboard - connections, building circuits



Ref: <https://www.allaboutcircuits.com/textbook/direct-current/chpt-5/building-simple-resistor-circuits/>

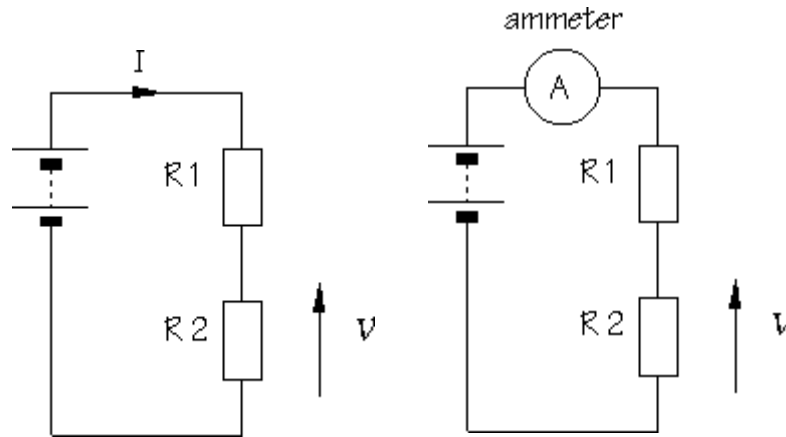
Task:

You have given four resistors of different values, which are R_1 , R_2 , R_3 and R_4 . Measure the value of the resistors.

1. Now, draw a circuit, where the resistors R_2 and R_3 are connected in parallel; and R_1 and R_4 are connected in series [$R_1 + (R_2 || R_3) + R_4$].
2. Implement the circuit in breadboard and measure equivalent resistance of the circuit.

Current Measurement

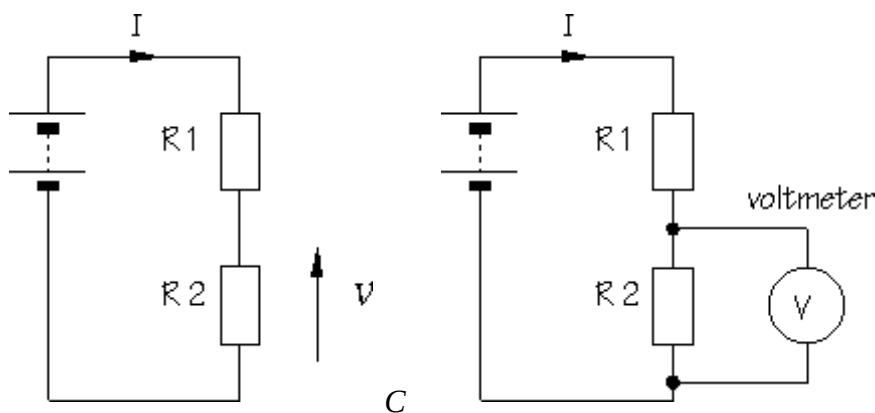
We have to make to a practical circuit in order to include the ammeter. To start with, you need to break the circuit so that the ammeter can be connected in series. All the current flowing in the circuit must pass through the ammeter. Meters are not supposed to alter the behavior of the circuit, or at least not significantly, and it follows that an ammeter must have a very LOW resistance.



To measure current, the circuit must be broken to allow the ammeter to be connected in series. Ammeters must have a LOW resistance

Voltage measurements:

This time, you do not need to break the circuit. The voltmeter is connected in parallel between the two points where the measurement is to be made. Since the voltmeter provides a parallel pathway, it should take as little current as possible. In other words, a voltmeter should have a very HIGH resistance.



To measure potential difference (voltage), the circuit is not changed: the voltmeter is connected in parallel. voltmeters must have a HIGH resistance

Observation: -

S no.	Supply Voltage	Resistance	Current	Voltage across resistor
1.	5V	R1=..... R2=.....		V _{R1} =..... V _{R2} =.....
2.	5V	R1=..... R2=.....		V _{R1} =..... V _{R2} =.....
3.	10V	R1=..... R2=.....		V _{R1} =..... V _{R2} =.....
4.	10V	R1=..... R2=.....		V _{R1} =..... V _{R2} =.....

Precautions:

- (a) The wires should be connected properly to the bread board.
- (b) Measurement should be taken properly.
- (c) Switch off the multimeter.

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Experiment No : 02

Name of the Experiment: Verification of Ohm's Law

Objective: To establish the voltage-current relationship in an electrical circuit

Theory:

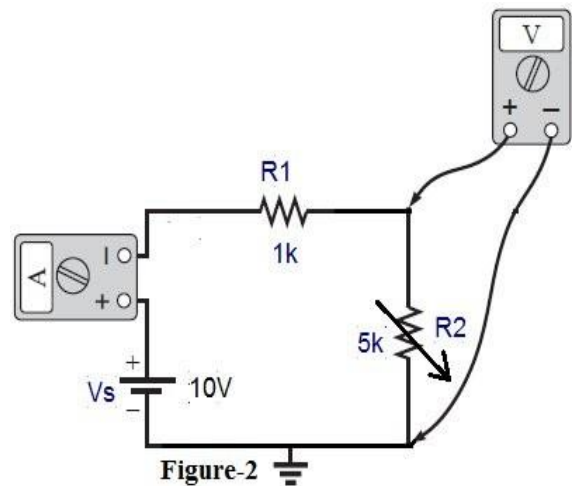
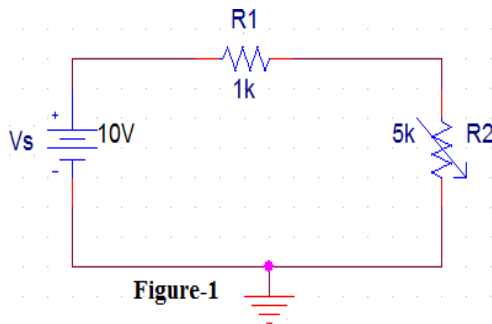
Ohm's law states that, in an electrical circuit, the current passing through a conductor between two points is proportional to the potential difference (i.e. voltage drop or voltage) across the two points, and inversely proportional to the resistance between them.

In mathematical terms, this is written as:

$$I = \frac{V}{R}$$

where I is the current in amperes, V is the potential difference in volts, and R is a constant, measured in ohms, called the resistance. The potential difference is also known as the voltage drop.

Circuit Diagram:



Apparatus:

1. Multimeter
2. DC voltage source
3. Resistors 1k-2, 2k-1, 3k-1, 4k-1, 5k-1

Procedure:

1. Make the circuit diagram as shown.
2. Keep the resistor R2 1k and R1 1k.
3. Take the reading of voltage across the R2 and Current through R2.
4. Change the value of R1 to 2k, 3k, 4k and 5k take the readings of voltage and current every time.
5. Plot the values of voltage and current and verify ohm's law.

Observation:

Voltage, V(V)					
Current, I (A)					

Report:

Draw the graphical representation of voltage-current relationship.

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Experiment No : 03

Name of the Experiment: Verification of Kirchhoff's Current Law

Objective: To verify Kirchhoff's Current Law (KCL) using nodal analysis of the given circuit.

Theory: Kirchhoff's Current Law states that the algebraic sum of all the currents meeting at any node (junction) is zero. In another way KCL implies that the total currents entering a node is equal to the total currents leaving the node.

Applying Kirchhoff's current law to the first four nodes in the circuit shown in Figure1 yields the following equations;

Node a: $-I_s + I_1 = 0$ (2a)

Node b: $-I_1 + I_2 + I_3 = 0$ (2b)

Node c: $-I_3 + I_4 = 0$ (2c)

Node d: $-I_2 - I_4 + I_5 = 0$ (2d)

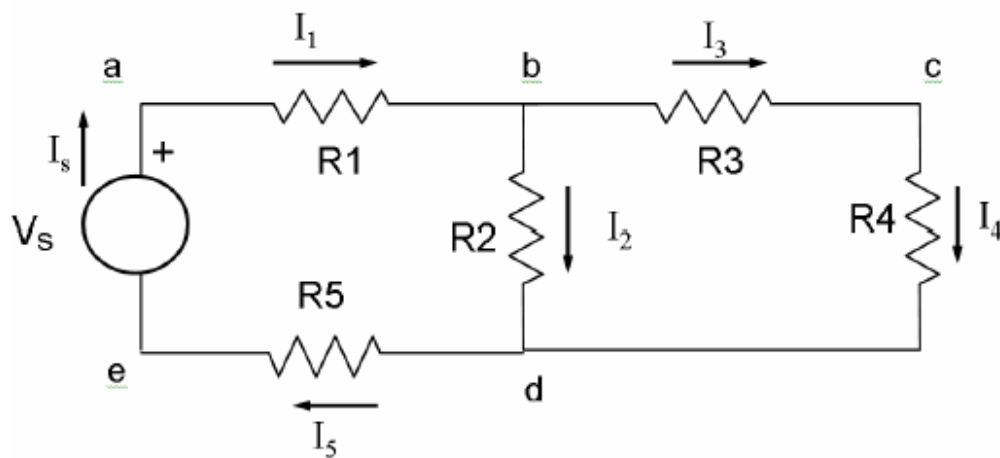


Figure 1

Apparatus:

1. Resistors (5)
2. DC source
3. Multimeter
4. Bread Board

Procedure:

1. Construct the circuit in bread board as shown in the figure 1.

$R_1 = 1\text{k}\Omega$

$R_2 = 2.4\text{k}\Omega$

$R_3 = 1.2\text{k}\Omega$

$R_4 = 1\text{k}\Omega$

$R_5 = 1\text{k}\Omega$

2. Set the dc power supply at 5 volts.

3. Measure the currents I_S , I_1 , I_2 , I_3 , I_4 and I_5 through the branches as shown in the figure with the help of the multimeter.

Observation:

Verify the equations 2a, 2b, 2c and 2d with the measured values of currents.

Reports:

1. Theoretically calculate the currents for each element in the circuit and compare them to the measured values.

2. Compute the percentage error in the two measurements and provide a brief explanation for the error.

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Experiment No : 04

Name of the Experiment: Verification of Kirchhoff's Voltage Law

Objective: To verify Kirchhoff's Voltage Law (KVL) using mesh analysis of the given circuit.

Theory: Kirchhoff's Voltage Law states that the algebraic sum of all the voltages around any closed path (loop or mesh) is zero. In other way the sum of the voltage drops around a closed path or loop is equal to the sum of the voltage rises around that path.

Applying Kirchhoff's voltage law to the first and the second loops in the circuit shown in Figure 1 yields:

Loop 1: $-V_s + V_1 + V_2 + V_5 = 0$ (1a)

Loop 2: $-V_2 + V_3 + V_4 = 0$ (1b)

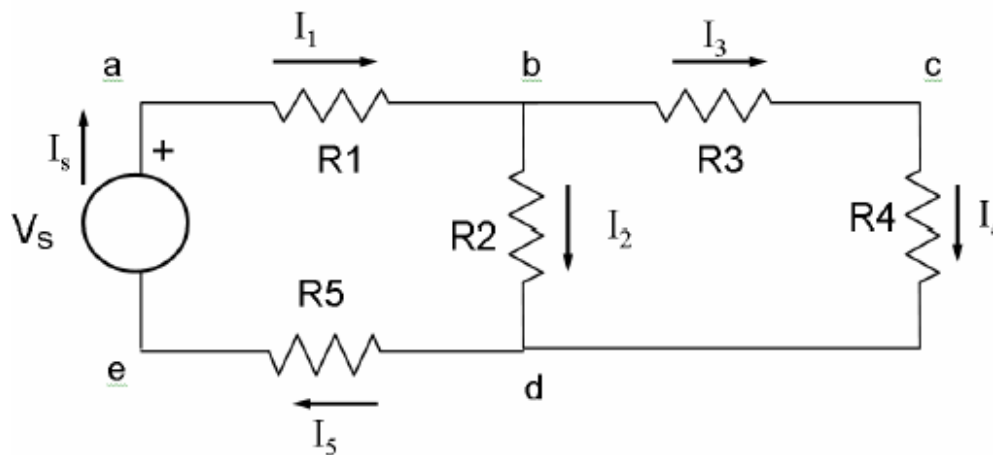


Figure 1

Apparatus:

1. Resistors (5)
2. DC source
3. Multimeter
4. Bread Board

Procedure:

1. Construct the circuit in bread board as shown in the figure 1.

$R_1 = 1.2\text{k}\Omega$, $R_2 = 3\text{k}\Omega$, $R_3 = 1\text{k}\Omega$, $R_4 = 3\text{k}\Omega$, $R_5 = 5.6\text{k}\Omega$

2. Set the dc power supply at 5 volts.

Measure the voltages across the resistors and note down the data in the table

3. Measure the total current.

4. Change the source voltage to 10 volts and repeat the instructions 2 to 4.

Observation:

No. of observation	Source voltage(V)	Voltage across (V)					Total current(A)
		R_1	R_2	R_3	R_4	R_5	

Calculation:

Verify whether your data agree with KVL.

Reports:

1. Theoretically calculate the voltages for each element in the circuit and compare them to the measured values.

2. Compute the percentage error in the two measurements and provide a brief explanation for the error.

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Experiment No : 05

Name of the Experiment: Study of Super Position Theorem in Circuit Analysis.

Objective:

To understand and study the superposition theorem.

Theory:

Superposition Theorem states that the response in a linear circuit with multiple sources can be obtained by adding the individual responses caused by the separate independent sources acting alone.

For an independent source acting alone, all other independent voltage sources are replaced by short circuits and all other independent current sources are replaced by open circuits.

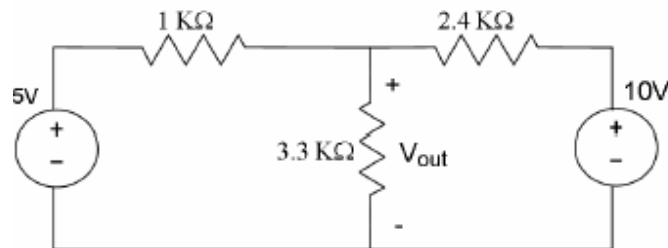


Figure 1

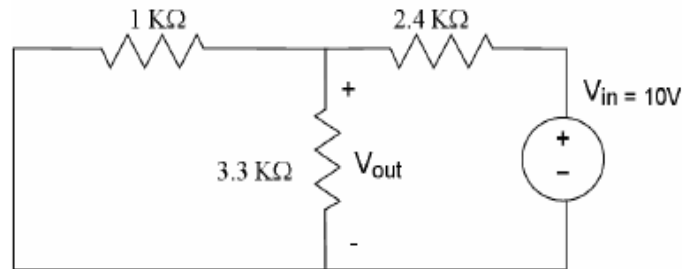


Figure 2

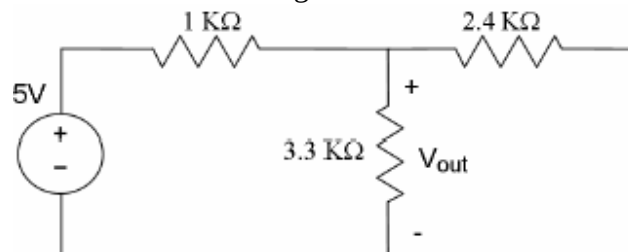


Figure 3

Apparatus:

1. Resistors (3)
2. Bread Board
3. Multimeter
4. DC power supply

Procedure:

1. Connect the three resistors R1, R2 and R3 with the two variable DC power supply as shown in Figure – 03.
2. Keep the both sources active.
3. Apply 20 volts (preferably) from E1 and 15 volts (preferably) from E2.
4. Measure the current I from the ammeter and record it in the given table.
5. Now, keeping E1 active and E2 inactive, measure the voltage from the multimeter.
6. Again keeping E2 active and E1 inactive, measure voltage from the multimeter.
7. Verify if $V = V' + V''$ WHICH would validate the superposition theorem for this particular circuit.
8. Repeat the steps 3 to change the values of E1 and E2 and record a few more sets of readings.

Experimental Data:

No. of Obs.	E ₁ (volts)	E ₂ (volts)	V (keeping E ₁ and E ₂ active) (amp)	V' (keeping E ₁ active and E ₂ inactive) (amp)	V'' (keeping E ₁ inactive and E ₂ active) (amp)
01.					
02.					
03.					

Reports:

1. Provide the results in tabular form.
2. Comment on the results and anomalies (if any).

Home Task:

Include the answers of the following questions in your lab report –

1. How can you make a voltage source inactive?
2. How can you make a current source inactive?
3. What are the benefits of this theorem?
4. Is it possible to apply Superposition theorem to nonlinear circuit?
5. Is it possible to apply Superposition theorem to ac as well as dc circuit?
6. How to find power using Superposition theorem?
7. Write some of the applications of the super position theorem.

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Experiment No : 06

Name of the Experiment: Study of Thevenin's Theorem in Circuit Analysis.

Objective: To verify Thevenin's theorem by obtaining the Thevenin equivalent voltage (V_{TH}) and Thevenin equivalent resistance (R_{TH}) for the given circuit.

Theory:

Thévenin's Theorem is a process by which a complex circuit is reduced to an equivalent series circuit consisting of a single voltage source (V_{TH}), a series resistance (R_{TH}) and a load resistance (R_L). After creating the Thévenin Equivalent Circuit, the load voltage (V_L) or the load current (I_L) may be easily determined.

One of the main uses of Thévenin's theorem is the replacement of a large part of a circuit, often a complicated and uninteresting part, by a very simple equivalent. The new simpler circuit enables us to make rapid calculations of the voltage, current, and power which the original circuit is able to deliver to a load. It also helps us to choose the best value of this load resistance for maximum power transfer.

Required Instruments:

- 1) Variable DC Power Supply
- 2) Multi-meter/voltmeter
- 3) Ammeter
- 4) Resistors
- 5) Connecting Wires
- 6) Bread board

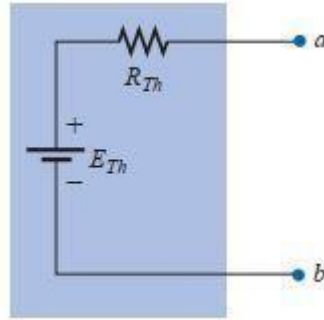


Figure – 1: Equivalent Circuit of the Thevenin's Theorem

V_{Th} = The open circuit voltage (V_{OC}) at the two terminals a and b.

R_{Th} = The resistance looking into the terminals A and B of the network with all sources removed.

There are several methods for determining Thevenin resistance R_{Th} . An attractive method for determining R_{Th} is:

1. Determine the open circuit voltage
2. Determine the short circuit current I_{SC} as shown in the figure

Circuit Diagram:

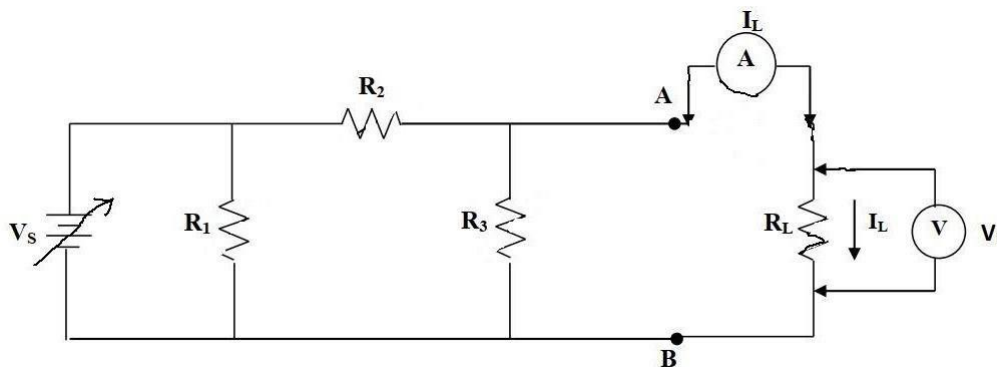


Figure – 3: Original Circuit Diagram

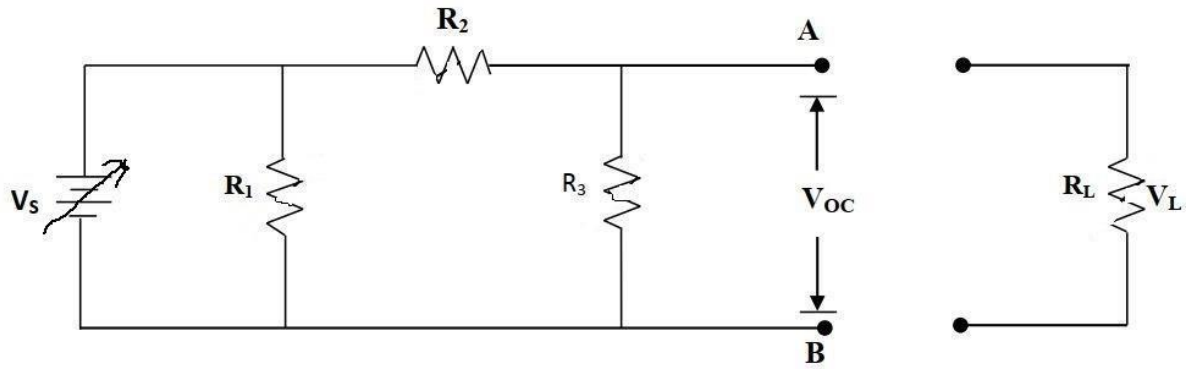


Figure – 4: Circuit Diagram for finding V_{oc}

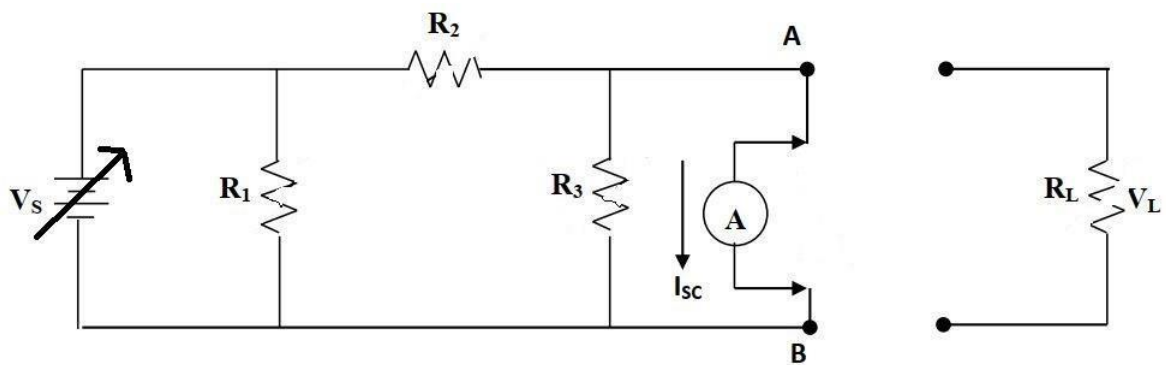


Figure – 5: Circuit for finding I_{sc}

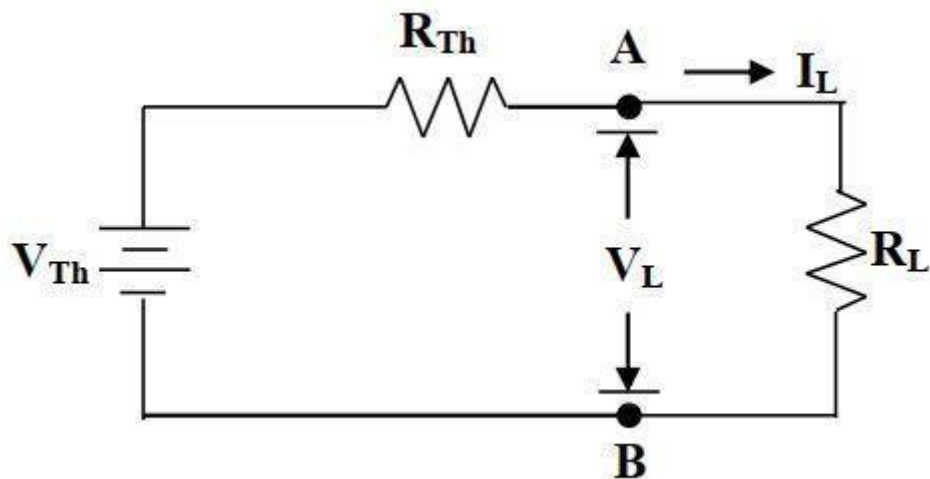


Figure – 6: Thevenin equivalent circuit for figure - 3

Steps to follow:

1. Connect the three resistors R_1 , R_2 and R_3 with a variable DC power supply as shown in Figure – 03.
2. Measure V_L and I_L for different R_L and record the data in Table.
3. Remove the load resistance R_L and find the open circuit voltage between terminals **A & B**. This voltage is Thevenin voltage i.e. $V_{Th}=V_{OC}$.
4. Place a short circuit between terminals **A & B** and find the short circuit current I_{SC} . Divide the open circuit voltage by the short circuit current to find the Thevenin resistance R_{Th} .
5. Construct the Thevenin's equivalent circuit as shown in Figure – 6, setting the powersupply at V_{Th} volts and the resistor at R_{Th} ohms. Now measure the load current I_L and the load voltage V_L for the values of R_L determined in step 2. Compare these values with previous values.

Experimental Data:

Table – 1: Data for Original Circuit

$R_1 =$ $R_2 =$ $R_3 =$ $V =$

No. of Obs.	R_L (ohms)	V_L (volts)	I_L (amps)
01.			
02.			
03.			
04.			
05.			

$V_{Th} =$ $R_{Th} =$

Table – 2: Data for Thevenin Circuit

No. of Obs.	R_L (ohms)	V_L (volts)	I_L (amps)
01.			
02.			

Reports:

1. Provide the results in tabular form.
2. Comment on the results and anomalies (if any).

Home Task:

Include the answers of the following questions in your lab report –

1. Define unilateral, bilateral & equivalent circuit.
2. Describe other methods for determining Thevenin resistance.
3. Mention the advantages of using Thevenin Theorem.

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Experiment:07 Familiarization with alternating current (AC) waves

Objective:

The Introduction to AC Circuit Lab is designed to give you hands-on experience in understanding the fundamental concepts of AC circuits. In this lab, you will learn how to measure AC voltage and current, how to calculate impedance and phase angle, and how to construct basic AC circuits using resistors, capacitors, and inductors.

- (a) Experimental observation of the Alternating Current (AC) waves.
- (b) Familiarization with the Function Generator and the Oscilloscope.

Required Instruments:

- 1) Function Generator
- 2) Oscilloscope
- 3) Resistor
- 4) Capacitor
- 5) Inductor
- 6) Bread board
- 7) Probes / Connecting wires

Generation of an alternating (AC) wave: An alternating wave (voltage or current) is an oscillating wave which alternately increases and decreases in magnitude with respect to time (Fig. 2). In an AC wave, the movement (or flow) of electric charge periodically reverses its direction. For instant, an electric charge moves forward and backward over and over again. Indirect current (DC), the movement (or flow) of electric charge is only in one direction.

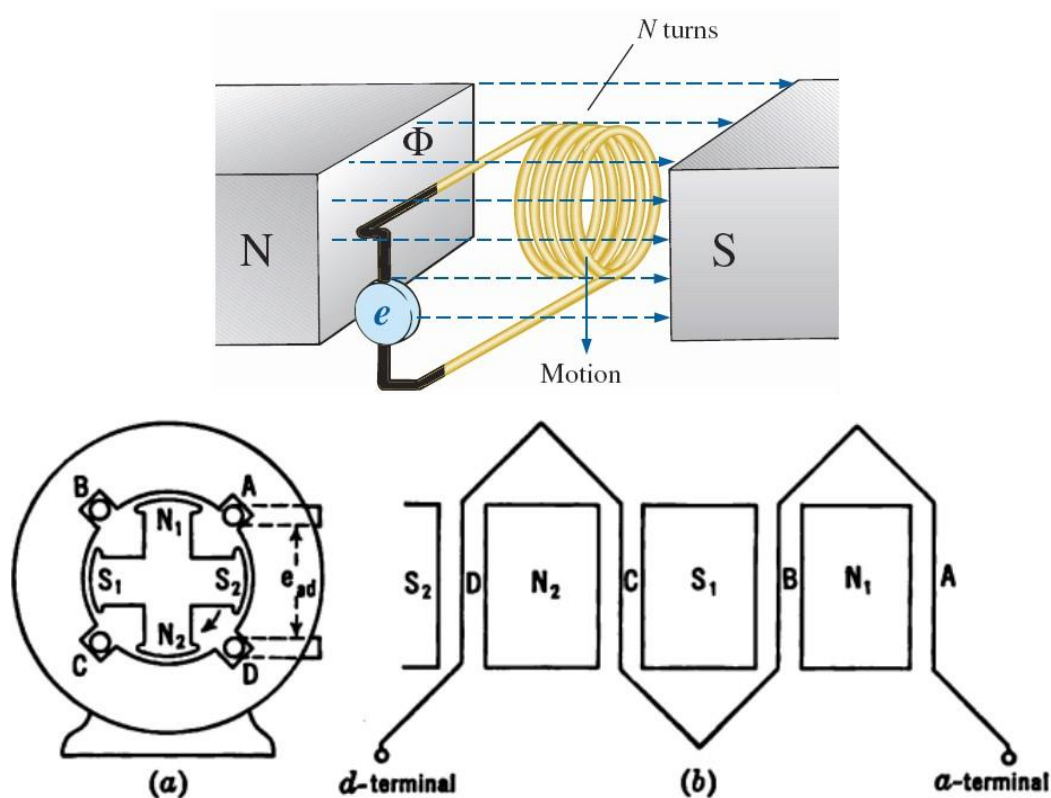


Fig. 1 Top image: Faraday's law of electromagnetic induction, Bottom image: A four pole, four conductor a-c generator of the revolving field type. The right in the bottom image shows the connection of the conductors.

According to the Faraday's law of electromagnetic induction (already described in the class EEE 111), when the magnets move across the conductors, a potential difference or *emf* is induced in the conductors.

$$e = N \frac{d\phi}{dt} \quad (\text{volts, V})$$

One can have a guess about the direction and magnitude of the induced *emf*. As shown in Fig.1 (bottom left), at the present condition, there would be "zero" *emf* developed across the conductors since no flux is being cut by the armature conductors. After one-eighth revolution of the magnets, the maximum *emf* will be developed and the voltage rise will be from the terminal *a* to terminal *d*. After quarter revolution, the *emf* will be zero again. The voltage will be maximum again and the voltage rise will be from the terminal *d* to terminal *a*. Therefore, the terminal *a* and terminal *d* of the generator would be alternatively positive and negative with respect to time and this time varying potential difference is shown in Fig. 2.

Voltage or Current wave form: The shape of the curve resulting from the plot of instantaneous values of voltage or current as ordinate against time as abscissa is its wave form or wave shape (Fig. 2).

Frequency: It is the number of voltage or current generating cycles per second.

Angular Velocity or Angular Frequency: The time to complete a cycle is defined as the period T . The angular velocity ω in radians per sec is $2\pi/T$, Therefore,

$$\omega = 2\pi/T = 2\pi f$$

Alternating Voltage or Current can be defined with any shape of waves. However, the usual trend is to define it by sine waves. Many of the AC waves in practice follow the trend of the sine waves, therefore, AC voltage and AC current calculations in the texts are based on the sinewaves. A true sine wave is expressed by the equation:

$$i = I_m \sin \omega t$$

where ωt is expressed in radians and is called the time angle, i is the instantaneous value of current, and I_m is the maximum value of the sinusoidal variation. Since ωt represents the angle, it can be expressed in radians or degrees, as follows-

$$i = I_m \sin \alpha$$

where α is expressed in radians or degrees. The physical meaning of this two equation is that, the former presents the current as a sinusoidal variation with respect to time, whereas the later represents the current as a sinusoidal variation with respect to angles.

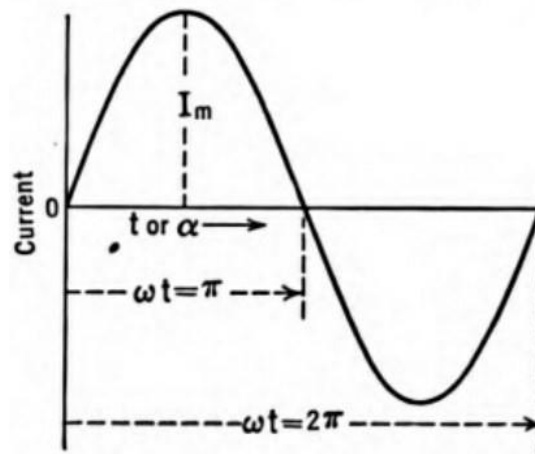


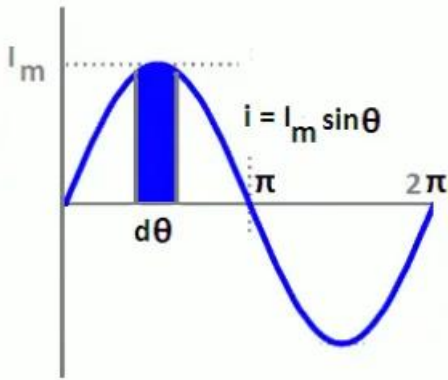
Fig. 2 A conventional schematic to represent a sine wave.

For the wave form in Fig. 2, Time

$$\text{period} = T \text{ Frequency } f = 1/T$$

$$v = V_m \sin 2\pi ft = V_m \sin(2\pi /T)t$$

Effective Value:



The term *rms* stands for (square) root of the mean of the squares of the instantaneous current values.

$$\begin{aligned}
 \therefore I_{rms} &= \sqrt{\frac{1}{\pi} \int_0^{\pi} i^2 \cdot d\theta} \\
 I_{rms}^2 &= \frac{1}{\pi} \int_0^{\pi} I_m^2 \sin^2 \theta \, d\theta = \frac{I_m^2}{\pi} \int_0^{\pi} \sin^2 \theta \cdot d\theta \\
 &= \frac{I_m^2}{\pi} \int_0^{\pi} \left(\frac{1 - \cos 2\theta}{2} \right) \cdot d\theta = \frac{I_m^2}{2\pi} \int_0^{\pi} (1 - \cos 2\theta) \cdot d\theta \\
 &= \frac{I_m^2}{2\pi} \left[\theta - \frac{1}{2} \sin 2\theta \right]_0^{\pi} \\
 &= \frac{I_m^2}{2\pi} \left[\left(\pi - \frac{1}{2} \sin 2\pi \right) - \left(0 - \frac{1}{2} \sin 2 \times 0 \right) \right] \\
 &= \frac{I_m^2}{2\pi} [\pi - 0 - 0 + 0] = \frac{\pi}{2\pi} I_m^2 = \frac{I_m^2}{2} \\
 \therefore I_{rms} &= \sqrt{\frac{I_m^2}{2}} = \frac{I_m}{\sqrt{2}} = 0.707 I_m
 \end{aligned}$$

These values are directly measured in AC voltmeter / ammeters and can be used in power calculation as:

$$P = I^2 R = V^2 / R$$

Phase: Phase is the fractional part of the period through which time or the associated time angle ωt has advanced from an arbitrary reference. The phase angle of a single wave is the angle from the zero point on the wave to the value at the point from which time is reckoned.

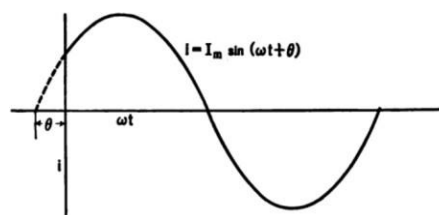


Fig. 3 Phase angle θ of a sine wave.

In Fig. 3, the equation $i = I_m \sin (\omega t + \theta)$ represents the sine wave of current with a phase angle of θ .

Phase difference: Phase difference between two AC sinusoidal waveforms is the difference in electrical angle between two identical points of the two waves. In Fig. 4, the voltage and current equations are given as:

$$v = V_m \sin \omega t$$

$$i = I_m \sin (\omega t + \theta)$$

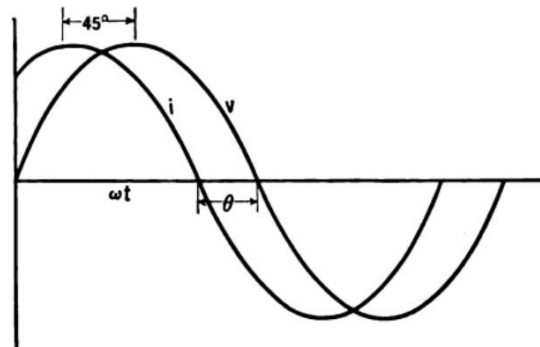


Fig. 4 A scenario where i wave leads the v wave by $\theta = 45^\circ$

Impedance: The relationship between the voltage across and the current through any component of an AC circuit is given by impedance. For the voltage and current waveforms in Fig. 4, the corresponding impedance Z is given as:

$$Z = V_m / I_m < \theta = V_{rms} / I_{rms} < \theta$$

Circuit Diagram:

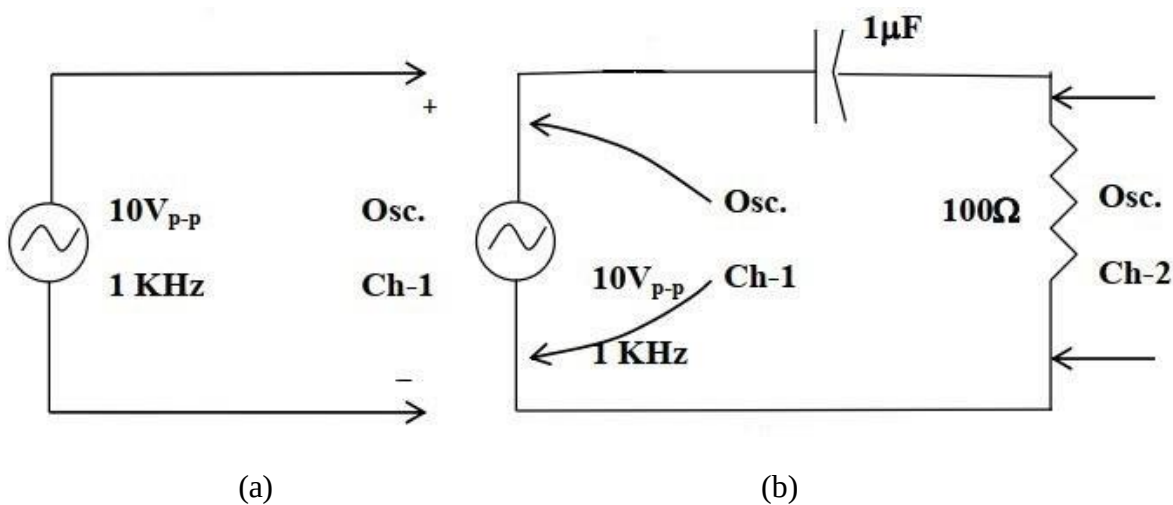


Fig. 5 Circuit diagram for the familiarization with the AC wave

Steps to follow:

1. Calibrate the Oscilloscope properly.
2. Connect the output of the function generator directly to channel 1 of the Oscilloscope as shown in Fig. 05 (a). Set the amplitude of the wave at 10 V_{p-p} and the frequency at 1 kHz. Select sinusoidal wave shape.
3. Determine the time period of the wave and calculate the frequency.
4. Measure the r.m.s voltage from the wave shape.
5. Change the frequency to 500 Hz and note what happens to the display of the wave. Repeat when the frequency is increased to 2 KHz.
6. Construct the circuit as shown in Fig. 5 (b). Measure the input voltage and the input current. The ratio between the voltage and the current gives the magnitude of the impedance, **Z**.
7. Observe the wave shapes of oscilloscope channels 1 and 2 simultaneously. Find the frequency of both the waves and amplitude from the display. Determine the phase difference between the two waves. The phase difference is given by $360 \frac{t}{T}$, where t is the time delay between the two waves. Also observe which of the two waves lead.

Note that the voltage in channel 2 is the voltage across a resistance and hence this is inphase with the current flowing in the circuit.

EXPERIMENT NO: 8

NAME OF THE EXPERIMENT : STUDY OF SERIES AND PARALLEL R-L-C CIRCUIT.

OBJECTIVE:

- i) To be familiar with ac quantities, their phase and phase differences.
- ii) As the ac quantities are vectors (Phasor), their additions, subtractions, and multiplications are in vector form. Hence, to be familiar with the vector diagrams drawing of a RLC series and a parallel circuits.

THEORY

The phase of an alternating quantity is very important to locate it properly with respect to a reference. Phase is the fractional part of a period through which time or the associated time angle ($\theta = \omega t$) has advanced from an arbitrary reference.

Fig. 9(a) shows phase relation between current through and voltage across resistive, purely inductive and purely capacitive elements.

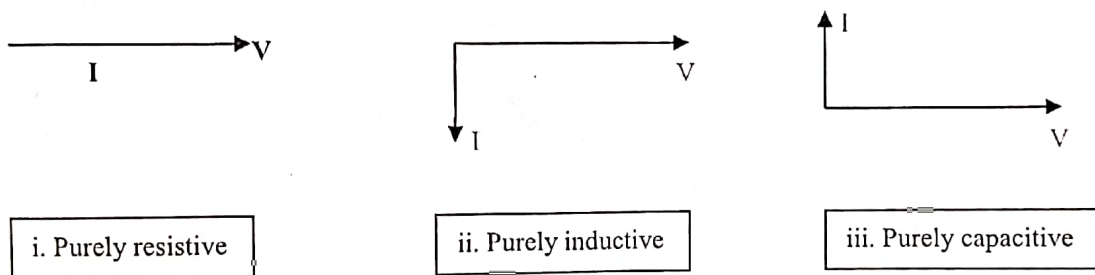


Fig. 9(a): Vector Diagram

In RLC series circuit, if a sinusoidal voltage is applied, then, according to KVL, the applied voltage must be equal to the vector addition of V_R , V_L , V_C .

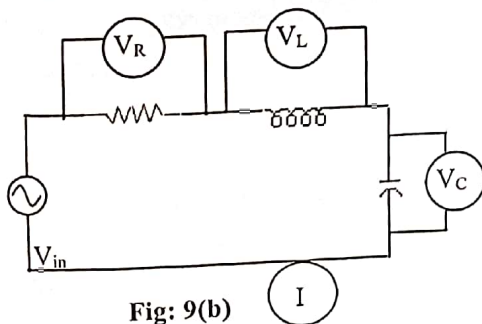


Fig: 9(b)

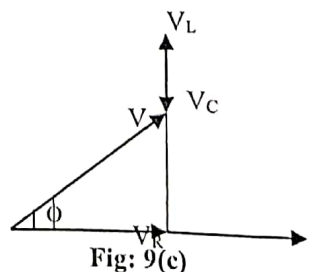


Fig: 9(c)

$$V_{in} = V_R + V_L + V_C$$

$$\Rightarrow V_{in} = IR + jIX_L - jIX_C$$

$$= I \{R + j(X_L - X_C)\}$$

$$\Rightarrow V_{in}/I = Z = \sqrt{R^2 + (X_L - X_C)^2} \angle \tan^{-1}(X_L - X_C)/R$$

Here, V and I indicate the rms values, Z is the impedance.

In a series circuit, current through the circuit is constant. Hence, to draw vector diagram of a series circuit, it is convenient to take current as reference. Conversely for parallel circuit voltage across parallel elements is taken as reference.

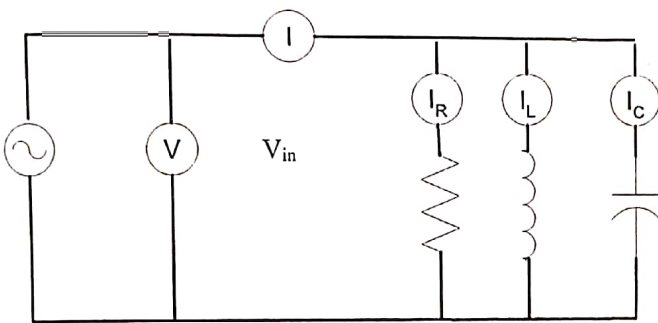


Fig: 9(d)

APPARATUS:

- Signal generator
- Capacitor
- Resistor
- Inductor
- Multimeter
- Bread board
- Oscilloscope
- Connecting wires etc.

PROCEDURE

- i. Implement the circuit of Fig. 9(b). Apply sinusoidal signal of 1000 Hz from the signal generator. Take the readings of I, voltages V_{in} , V_R , V_L , V_C using multimeter.
- ii. Then draw a vector diagram with the help of the method shown in Fig. 9(c).
- iii. Similarly, connect the parallel circuit as in Fig. 9(e) and take the readings of V_{in} , V, I, I_R , I_L , I_C . Then draw a vector diagram following the hint given.

DATA TABLE:

(a) For series circuit:

V (in) volt	I ma	V _R volt	V _L volt	V _C volt

(b) For parallel circuit :

f Hz	V (in) volt	V volt	I ma	I _R ma	I _L ma	I _C ma

REPORT:

1. Show the table.
2. Comment on the obtained results and discrepancies (if any).

EXPERIMENT NO. 9

Experiment Name- Study the characteristics of High Pass Filter.

Objective-

The purpose of this experiment is to study the characteristics of high pass filter. Observe the input-output curve of the High pass filter. Learn the working procedure of the high filter.

Theory

Electronic filters are circuits which perform signal processing functions, specifically to remove unwanted frequency components from the signal, to enhance wanted ones, or both.

A **high-pass filter** is a filter that passes high frequencies well, but attenuates (or reduces) frequencies lower than the cutoff frequency. The actual amount of attenuation for each frequency varies from filter to filter. A high-pass filter is the opposite of a low-pass filter. It is useful as a filter to block any unwanted low frequency components of a complex signal while passing the higher frequencies.

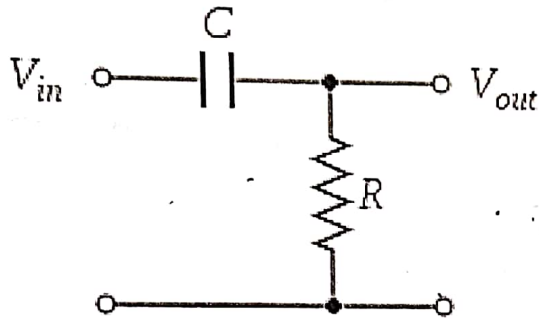
Cut off frequency: electrical engineering, a **cutoff frequency**, **corner frequency**, or **break frequency** is a boundary in a system's frequency response at which energy flowing through the system begins to be reduced (attenuated or reflected) rather than passing through.

Equipment:

1. Capacitor.
2. Resistor.
3. Function Generator.
4. Oscilloscope.

Work Procedure:

1. Connect the circuit as shown below.
2. Use a capacitor and Resistors
3. Give the input from power supply.
4. Connect Oscilloscope.
5. Observe input wave shape at channel 1 and output wave at channel 2

Circuit Diagram:**Lab Report:**

1. Observe the variation of output curve with the change of input frequency.
2. Take all the data.
3. Draw the Necessary graph. (Use graph paper).
4. Make comment on every input-output graph.

Result:**Discussion:**

EXPERIMENT NO. 10

Experiment Name- Study the characteristics of Low Pass Filter.

Objective-

1. The purpose of this experiment is to study the characteristics of low pass filter.
2. To observe the input-output curve of the low pass filter.
3. To plot output voltage-frequency response of low pass filter.

Theory

Electronic filters are circuits which perform signal processing functions, specifically to remove unwanted frequency components from the signal, to enhance wanted ones, or both.

A **low-pass filter** is a **filter** that **passes** signals with a frequency **lower** than a certain cutoff frequency and attenuates signals with frequencies higher than the cutoff frequency. A **low-pass filter** is the complement of a **high-pass filter**.

Cut off frequency: electrical engineering, a **cutoff frequency**, **corner frequency**, or **break frequency** is a boundary in a system's **frequency** response at which energy flowing through the system begins to be reduced (attenuated or reflected) rather than passing through.

Equipment's:

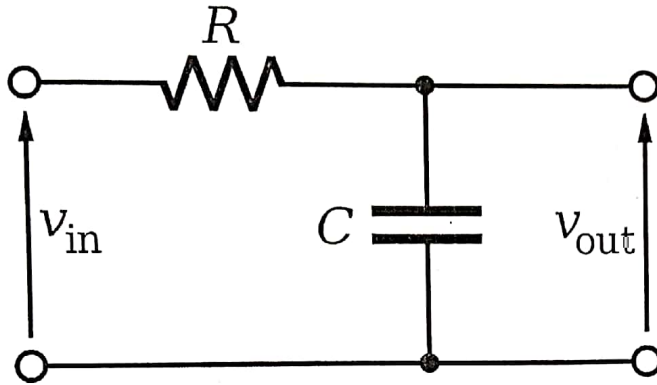
1. (10 PF) Capacitor.
2. 10 k Resistor.
3. Function Generator.
4. Oscilloscope.
5. Circuit Breadboard

Work Procedure:

1. Connect the circuit as shown below.
2. Use a capacitor and Resistors
3. Give the input from power supply.
4. Connect Oscilloscope.

5. Observe input wave shape at channel 1 and output wave at channel 2

Circuit Diagram:



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Lab Report:

1. Observe the variation of output curve with the change of input frequency.
2. Take all the data.
3. Draw the Necessary graph. (Use graph paper).
4. Make comment on every input-output graph.

Result:

Discussion: